

Mohammad Taiyab Khan

Data & AI Engineer

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Portfolio: taiyab.autostrata.ai · Egham, Surrey, UK · Right to work in the UK (Graduate Visa to Jan 2027)

PROFESSIONAL SUMMARY

Data & AI Engineer with an MSc in Data Science & Analytics (Merit) from Royal Holloway, University of London. I build AI products that move from idea to demo to deployed: a live agentic AI assistant anyone can try at taiyab.autostrata.ai, a face-verified attendance product heading to market, and shipped features on live client platforms including a Stripe deposit flow on a rental marketplace. Founding engineer at Autostrata, a two-person AI product studio, where I design, build, and run our production systems end-to-end. Python-first across ML, data, and APIs, with production experience in TypeScript, React/Next.js, and Node in an AI-assisted workflow; equally comfortable presenting findings to non-technical stakeholders.

PROFESSIONAL EXPERIENCE

Founding AI Engineer | Autostrata

May 2025 – Present · Egham, UK

Two-person AI product studio; I lead all engineering — design, build, and operation of our production systems.

- Built and shipped a production agentic AI assistant, live at taiyab.autostrata.ai — Vercel AI SDK with Claude (Haiku 4.5 via OpenRouter, ~\$0.005/turn), strict routing across three tools (closed-knowledge Q&A, Cal.com API v2 for real in-chat calendar bookings, Resend email), hard refusal patterns, per-IP rate limiting, and streaming responses.
- Designed and built Clock-in Pro, a face-verified, GPS-geofenced attendance product for small businesses, live pre-launch at clockinpro.autostrata.ai — React/TypeScript front end, Node/Express (TypeScript) API, PostgreSQL (Neon serverless) with Drizzle ORM, server-side face-embedding matching (face-api.js / TensorFlow.js), deployed on Railway. Originated as a Python/DeepFace prototype I built hands-on; now being productised to sell to small businesses.
- Built MummyHelp end-to-end, a commissioned voice-activated SOS app prototype — React Native/Expo, Node/Express with Supabase (PostgreSQL), on-device wake-word detection (Picovoice Porcupine), and a Python Flask speech service (Vosk with Google Speech fallback); SOS alerts reached the paired device in ~3–4 s in device testing. Evaluated speaker-verification approaches (incl. SpeechBrain ECAPA-TDNN) before choosing on-device detection for latency and privacy.
- Commissioned by an academic researcher to build HEART-Eco, live at heart-eco.vercel.app — a Next.js 14 dashboard for a multi-pillar economic scoring model with 2026–2030 country forecasts, Recharts visualisations, and an OpenAI Assistants 'Ask AI' panel for natural-language exploration.
- Extended a live Sharetribe rental marketplace (Sartorial London) across two paid phases — Stripe security-deposit / charge-later flow, rebuilt 5-day booking logic, end-to-end shipping fees, and custom transaction states; the client returned for phase two and left a public review.
- Built the MVP of a SKU-level textile-provenance platform for an early-stage ethical-fashion startup — Next.js 14 + TypeScript, Supabase (Postgres, Auth), multi-tenant artisan-to-SKU data model with public QR provenance pages.
- Delivered additional small-business web builds (Clarivance AI, Empire Exchange, and others).

AI Content Analyst | Outlier (Contract)

Jan 2024 – Mar 2024 · London, UK

- Evaluated LLM-generated outputs for factual accuracy, domain relevance, and quality across STEM content pipelines — flagging errors, diagnosing root causes, and writing structured corrections for RLHF-style training workflows.
- Worked primarily in Python; assessed mathematical reasoning, code, and natural-language responses against quality rubrics.

Data Analyst (Part-time) | Gurucool

Jan 2022 – Aug 2023 · New Delhi, India

Part-time alongside my BSc, at an early-stage startup whose app reached ~50,000 MAU within months of launch.

- Built a Python hybrid recommendation system (collaborative + content-based filtering) for the BITS short-form content feature; co-authored an internal research paper on the methodology.
- Integrated the OpenAI GPT API (Davinci to GPT-3.5 Turbo) with custom prompt engineering for an in-app Q&A chatbot, built in the early ChatGPT era.
- Built Power BI MIS dashboards for weekly senior-management reviews; ran user-behaviour analysis (Google Analytics, GTM, Microsoft Clarity, Hotjar) that informed the Padhaai app UI redesign.
- Built a BeautifulSoup web scraper structuring data on 10,000+ Indian colleges to support content targeting.

Retail Assistant (Part-time) | Primark

Oct 2023 – Present · Camberley, UK

- Part-time alongside MSc and Autostrata; cash-office reconciliation and floor operations; Employee of the Quarter.

EDUCATION

MSc Data Science & Analytics — Merit | Royal Holloway, University of London

Sep 2023 – Sep 2024

- Modules: Machine Learning, Online Machine Learning, Large-Scale Data Storage & Processing (Spark, HDFS), Data Visualisation, Databases, Ethical Issues in AI.

- Dissertation (supervised by Prof. Daniel O’Keeffe): Predictive Power of Social Media — published as a preprint on Zenodo (see Projects).
- Diploma in Data Science — First Division (7.6 CGPA) | IIT Madras

Mar 2021 – Mar 2023
- BSc Computer Science & Mathematics — First Division (8.28 CGPA) | Jamia Millia Islamia

Sep 2020 – Sep 2023

SELECTED PROJECTS

- Predicting EURO 2024 via Social Media Sentiment — MSc Dissertation

DOI: 10.5281/zenodo.19675434
- Built four classifiers — TF-IDF + Logistic Regression (86% accuracy, AUC 0.927), VADER, three-class LR, and a zero-shot Claude Haiku classifier with a football-aware prompt; only the LLM detected the genuine win-day vs loss-day sentiment shift ($t = 2.52$, $p = 0.012$).
 - Key insight: 79% of football tweets are factually neutral, so benchmark accuracy and domain-level sensitivity are dissociable properties of classifiers. Full code, data, and paper public on GitHub; preprint on Zenodo.
- IronTrack — personal workout tracker

Next.js + Supabase
- Web app logging workouts (weight, reps, RPE) — one of the personal tools I build for everything I do (training, diet, expenses, investments).
- UK Visa Sponsorship Companies Map

Flask + Leaflet.js
- Interactive map of UK sponsor-licence companies with marker clustering, town/city filtering, and Google Maps geocoding.
- Earlier academic work: Algerian forest-fire risk app (Flask ML pipeline, Ridge Regression, deployed on Render); Alzheimer’s detection from MRI (84% accuracy, F1 0.8); e-commerce purchase-intent ensemble (81% accuracy, IIT Madras capstone); Cyclistic bike-share analysis (5.78M rides, SQL/Power BI); Enron email network analysis (Spark, HDFS).

TECHNICAL SKILLS

- Programming:** Python (pandas, NumPy, scikit-learn), SQL, TypeScript/JavaScript, R (academic)
- AI & LLM engineering:** Vercel AI SDK, Claude API (OpenRouter), OpenAI Assistants API, prompt engineering, RAG, agentic tool routing, n8n
- Machine learning & statistics:** Classification, regression, clustering, ensembles (XGBoost, Random Forest), statistical inference, A/B testing, time-series analysis
- Web & backend:** React 18, Next.js 14, Node.js/Express, FastAPI, Flask, REST APIs, Tailwind CSS, React Native (Expo), Stripe
- Data & infrastructure:** PostgreSQL (Neon), Supabase, Drizzle ORM, Docker, Railway, Vercel, GCP (basic), AWS (basic), Spark/HDFS (academic), BeautifulSoup
- Speech & vision:** Picovoice Porcupine, Vosk, face-api.js (TensorFlow.js)
- BI & analytics:** Power BI, Tableau, Google Analytics, GTM, Microsoft Clarity, Hotjar, Advanced Excel
- AI-assisted engineering:** Claude Code, Cursor — production development workflow
- Languages:** English (fluent), Hindi (native), Urdu (native)

CERTIFICATIONS

- Google Data Analytics Professional Certificate — Coursera, Feb 2024 · Machine Learning — Stanford University (Andrew Ng), May 2023 · Foundational Certificate in Data Science — IIT Madras

PUBLICATIONS

- Khan, M. T. (2026). Predictive Power of Social Media: A Comparative Study of Classical ML, Lexicon-Based, and LLM Approaches for Football Sentiment Analysis. Preprint, Zenodo. DOI: 10.5281/zenodo.19675434